

Likelihood-Ranked Synthetic Selection in Self-Consuming Diffusion: Minority Extinction Ordering at Moderate Imbalance

Muhammad Alamzeb
Independent researcher
shayankhanmahar@gmail.com

Abstract

Self-consuming training loops—retraining generative models on their own outputs—can induce *model collapse*, including loss of rare modes. We study unlabeled selection of synthetic samples by a denoising likelihood proxy in self-consuming DDPM training on imbalanced mixtures, with emphasis on **moderate** imbalance. Define the minority-mass ratio $r = m^{(1)}/m^{(0)}$. On a primary matrix of eight seed–imbalance cells, we report results **stratified by imbalance** ρ (not a single pooled test as the headline claim). At $\rho=5$ ($n=5$ seeds), generation 1 minority mass obeys bottom $>$ rand $>$ top on all seeds; paired retention contrasts vs. random- k have large effects (top–rand: Cohen’s $d_z=-1.96$, bootstrap 95% CI $[-0.37, -0.15]$; bottom–rand: $d_z=+1.65$, CI $[0.16, 0.43]$). Exact Wilcoxon signed-rank $p=0.0625$ for both (the minimum two-sided exact p achievable at $n=5$ when all signs agree). At $\rho=10$ ($n=3$), the full order holds on 0/3 seeds and contrasts are near null. A hypothesized “false stability” pattern—top- k improving sliced W_2 —is not observed at generation 1 ($\Delta W_2 > 0$ on 8/8 cells; exact Wilcoxon $p=0.0078$; Holm-adjusted across five generations $p=0.039$). Directional transfers (8D GMM, Digits-PCA; 3 seeds) match the $\rho=5$ order. Code: <https://github.com/muhammadalamzeb/likelihood-rank-self-consuming-diffusion>.

1 Introduction

As synthetic data proliferates, generative models are increasingly trained on mixtures of real and model-generated samples. Iterated self-consumption can cause *model collapse* [Shumailov et al., 2023, Alemohammad et al., 2023, Gerstgrasser et al., 2024]. Mitigations include accumulating real data [Gerstgrasser et al., 2024] and filtering synthetics [Feng et al., 2024, Cai et al., 2025]. When the only score is the generator’s own likelihood, how should one select synthetics? We isolate an *interventional* selection policy at matched synthetic count and matched real-mix α , with mode-exact metrics.

What is new. A controlled top- k / random- k / bottom- k contrast under matched budgets—not a new named training algorithm or ImageNet-scale SOTA method.

Contributions. (1) Matched-budget DDPM protocol with exact mode masses. (2) Likelihood-rank ordering of minority retention, **stratified by** ρ . (3) Evidence that top- k does not improve sliced W_2 vs. random- k . (4) Directional 8D GMM and Digits-PCA checks.

Table 1: Primary cells. Seeds 3–4 only at $\rho=5$.

$\rho \backslash$ seed	0	1	2	3	4
5	✓	✓	✓	✓	✓
10	✓	✓	✓	—	—

2 Related work

Shumailov et al. [2023], Alemohammad et al. [2023] document collapse and quality-biased sampling (*MAD*). Gerstgrasser et al. [2024] show accumulating real+synthetic data can avoid collapse. Feng et al. [2024] study task verification of labeled synth; Cai et al. [2025] (*LSF*) filter via latent probes. Inference-time temperature methods [Xu et al., 2025, Li et al., 2026] reshape sampling diversity; we intervene on training-set selection.

3 Method

3.1 Self-consuming loop

Train DDPM $f^{(g)}$ on $\mathcal{D}^{(g)}$, sample synthetics, mix fixed real buffer (fraction α) with selected synthetics.

3.2 Likelihood proxy and policies

Score $s(x)$: mean denoising MSE over a t -grid (lower = higher likelihood).

- **top- k / bottom- k / random- k :** pool of size $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$ by lowest / highest / uniform scores; then draw n_{syn} and mix with α real.
- **mix:** same α , but synthetic portion is a uniform subsample of all N synthetics (no rank filter).
- **replace:** pure synthetic.

Thus mix $\neq$ random- k .

3.3 Metrics

Nearest-mean mode masses; minority-mean m ; ratio $r = m^{(1)}/m^{(0)}$ (“retention,” allowed > 1). Sliced W_2 : mean squared 1D Wasserstein over **24** projections; ≤ 400 synth and ≤ 400 held-out real points. Hypothesis tests use the **exact** Wilcoxon signed-rank distribution over sign assignments (enumeration; midranks for magnitude ties), not only a normal approximation.

4 Experiments

4.1 Primary design ($n=8$ cells)

$K=4$, $G=5$, $\alpha=0.5$, $k_{\text{frac}}=0.5$. Logs: `w2_fs.jsonl`.

Table 2: Per- ρ mean r and seed-wise counts (generation 1).

ρ	n	bottom	rand	top	Full order	top<rand / bot>rand
5	5	0.737	0.456	0.183	5/5	5/5 / 5/5
10	3	0.296	0.199	0.154	0/3	2/3 / 1/3

Table 3: Stratified paired retention contrasts vs. random- k (exact Wilcoxon; bootstrap 95% CI of mean Δ).

ρ	Contrast	Mean Δ	d_z	95% CI	Exact p
5	top–rand	−0.273	−1.96	[−0.37, −0.15]	0.0625
5	bottom–rand	+0.281	+1.65	[0.16, 0.43]	0.0625
10	top–rand	−0.045	−0.28	[−0.19, 0.13]	0.75
10	bottom–rand	+0.097	+0.38	[−0.08, 0.39]	1.00

4.2 Retention: stratified primary analysis

Because $\rho=5$ and $\rho=10$ behave differently, we do **not** treat a single pooled Wilcoxon as the primary claim. Table 2 gives means and seed-wise order counts; Table 3 gives stratified paired contrasts.

At $\rho=5$, all five seeds agree in sign; exact two-sided $p=0.0625$ is the smallest attainable at $n=5$ under the signed-rank null when all signs agree (so lack of $p<0.05$ is a discrete- n limit, not a null effect— $|d_z|>1.6$). At $\rho=10$, minorities are near floor at $g=0$ and contrasts are consistent with null. For completeness, a pooled $n=8$ exact Wilcoxon yields $p=0.039$ for both contrasts, but we treat that as secondary because it mixes heterogeneous strata.

4.3 Sliced W_2

Generation-wise tests are repeated measures on the same cells (serially correlated), so Holm is a conservative acknowledgment rather than an assumption of independence. Evidence against top- k improving W_2 is strongest at $g=1$.

4.4 Real-mix α ablation (illustrative; $n=2$)

At $\alpha=0.5$, seed 0 breaks the full order (bottom $m^{(1)}$ slightly below random) while seed 1 holds—hence 1/2 (means still rank bottom>rand>top). At $\alpha=0.25$, likewise 1/2; at $\alpha=0.75$, top vs. random reverses on average (0/2).

4.5 Transfer checks (directional; $n=3$)

Minimum two-sided exact Wilcoxon p at $n=3$ with all signs agreeing is 0.25; we report Cohen’s d_z against that bound.

- **8D GMM:** 0.55 ± 0.20 / 0.27 ± 0.06 / 0.08 ± 0.03 ; order 3/3; top–rand $d_z=-2.56$, bottom–rand $d_z=+1.28$ (exact $p=0.25$).
- **Digits-PCA:** 1.49 ± 0.11 / 1.11 ± 0.08 / 0.68 ± 0.17 ; order 3/3; top–rand $d_z=-1.72$, bottom–rand $d_z=+2.67$ (exact $p=0.25$).

Primary retention by imbalance (stratified)

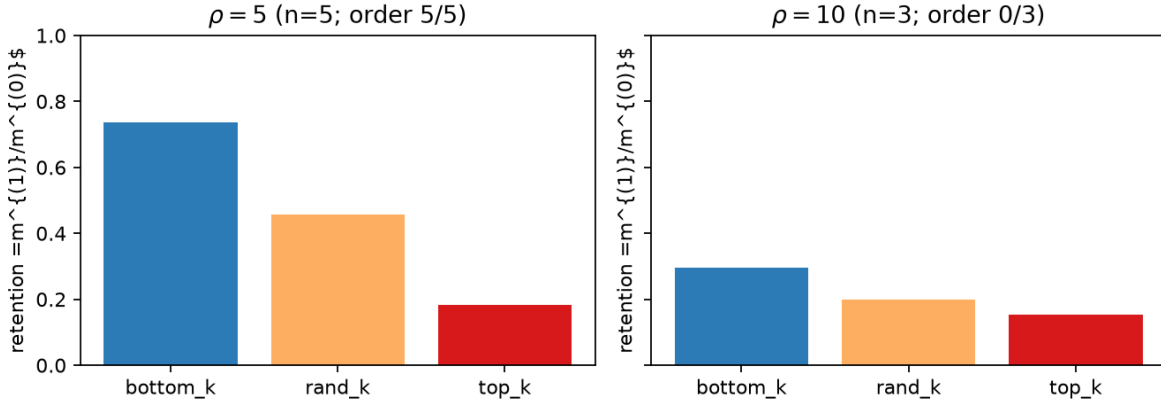


Figure 1: Primary retention r stratified by ρ (matches the primary claim). A pooled bar chart would mix the 5/5 and 0/3 regimes and is not used as the lead figure.

Table 4: $\Delta W_2(\text{top} - \text{rand})$ by generation on all eight cells (positive = top worse). Exact Wilcoxon p ; Holm adjustment across the five (non-independent) generation-wise tests; bootstrap 95% CI of the mean.

g	Mean	Std	d_z	95% CI	$\#\{> 0\}$	Exact p (Holm)
1	+0.113	0.099	1.14	[0.05, 0.18]	8/8	0.0078 (0.039)
2	+0.061	0.060	1.00	[0.02, 0.10]	6/8	0.039 (0.094)
3	+0.039	0.035	1.11	[0.02, 0.06]	7/8	0.023 (0.094)
4	+0.027	0.032	0.83	[0.01, 0.05]	7/8	0.023 (0.094)
5	+0.024	0.033	0.73	[0.01, 0.05]	7/8	0.023 (0.094)

4.6 Selection-pool size k_{frac} (illustrative; $n=2$)

Top- k remains worst on means; at 0.75, bottom vs. random flips (0/2 full order).

5 Analysis

Top- k retains high-likelihood (typically majority) synthetics, reducing minority inflow. The ordering is clear when minorities remain measurable ($\rho=5$) and near noise at $\rho=10$. Sliced W_2 is a poor monitor of minority survival here.

6 Limitations

Toy GMMs / Digits-PCA; not ImageNet-scale. At $\rho=5$, exact Wilcoxon cannot beat $p=0.0625$ with $n=5$ all-agreeing signs—inference leans on d_z and unanimous direction. α and k_{frac} ablations use two seeds; their default rows are the same experimental cell. Transfers: $n=3$ cannot beat exact $p=0.25$. **Multiplicity:** Holm is applied within families (two $\rho=5$ retention contrasts; five W_2 generations). Additional stratified/exploratory tests are not multiplicity-corrected paper-wide; we treat them as descriptive under the small-sample framing. Contribution is a controlled empirical

Table 5: Mean r by α (seeds $\{0, 1\}$, $\rho=5$, default $k_{\text{frac}}=0.5$). “Order holds” = bottom>rand>top on $g=1$ minority mass. The $\alpha=0.5$ row is the *same cell* as $k_{\text{frac}}=0.5$ in Table 6.

α	bottom	rand	top	Order holds
0.25	0.25	0.17	0.05	1/2
0.50	0.62	0.49	0.24	1/2
0.75	1.34	0.77	0.94	0/2

Table 6: k_{frac} ablation (seeds $\{0, 1\}$, $\rho=5$, $\alpha=0.5$). $0.25 \equiv 0.5$ when n_{syn} binds. The $k_{\text{frac}}=0.5$ row duplicates Table 5’s $\alpha=0.5$ row.

k_{frac}	bottom	rand	top	Order holds
0.25	0.62	0.49	0.24	1/2
0.50	0.62	0.49	0.24	1/2
0.75	0.48	0.61	0.22	0/2

contrast relative to MAD/LSF/Feng.

Note on double-blind review. The public GitHub URL reveals author identity. For double-blind venues, provide an anonymous code mirror; for arXiv, the named repository is appropriate.

7 Conclusion

Likelihood ranking induces a minority-retention order that is stable at **moderate** imbalance ($\rho=5$) and not seed-wise reliable at $\rho=10$ in our matrix. Top- k does not improve sliced W_2 vs. random- k at generation 1. Preferring “most likely” synthetics is not a free lunch for minority survival when rare modes remain measurable.

References

- Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoobi, and Richard G. Baraniuk. Self-consuming generative models go MAD, 2023.
- Zhongteng Cai, Yaxuan Wang, Yang Liu, and Xueru Zhang. Stabilizing self-consuming diffusion models with latent space filtering, 2025.
- Yunzhen Feng, Elvis Dohmatob, Pu Yang, Francois Charton, and Julia Kempe. Beyond model collapse: Scaling up with synthesized data requires verification, 2024.
- Matthias Gerstgrasser, Rylan Schaeffer, Apratim Dey, Rafael Rafailov, Dhruv Pai, Henry Sleight, John Hughes, Tomasz Korbak, Rajashree Agrawal, Andrey Gromov, Daniel A. Roberts, Diyi Yang, David Donoho, and Sanmi Koyejo. Is model collapse inevitable? breaking the curse of recursion by accumulating real and synthetic data, 2024.
- Peizhuo Li, Emre Aksan, Alexandru-Eugen Ichim, Thabo Beeler, and Olga Sorkine-Hornung. Diversify diffusion with temperature sampling and variance-corrective time shifting, 2026.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. The curse of recursion: Training on generated data makes models forget, 2023.

Yanbo Xu, Yu Wu, Sungjae Park, Zhizhou Zhou, and Shubham Tulsiani. Temporal score rescaling for temperature sampling in diffusion and flow models, 2025.

A Reproducibility

`python scripts/reproduce_w2.py`; stats tables via `experiments/scripts/make_submission_stats.py`.
Code: <https://github.com/muhammadalahzeb/likelihood-rank-self-consuming-diffusion>.